

Automated Assessment of Consent
Dialogues for GDPR Compliance

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Literature Review: Automated Assessment of Consent Dialogues for GDPR Compliance

1. Introduction

Consent dialogues are normally used on pages in order to comply with the GPDR and ePrivacy Directive by allowing users to decide how their information is handled. Even though the GDPR does not ask for the dialogues to be in place, it does clearly state in Article 7 that users of that data must provide their consent in an informed, specific, and unequivocal manner. Consent dialogues, whether in the form of cookie banners or opt-in forms, are now seen as an easy way for organizations to comply with the regulations. Even though, their effectiveness and compliance have been put into question. “*Research by Nouwens et al. (2020) states that approximately 93% of EU websites use dark patterns, which are defined as designs that push users to make non-compliant consent decisions.*”[\[1\]](#) These actions violate GDPR’s core requirements of ensuring transparency and obtaining consent.

Even though tools for privacy compliance creation, like the transformer-based NLP models (BERT) for privacy policy parsing, have evolved (Zhang et al., 2024), automatic assessment of the dynamic consent dialogue systems is still a largely untouched domain. Some of the main issues are:

1. **Structural heterogeneity:** Consent interfaces vary widely in design, terminology, and legal granularity (Nouwens et al., 2020)[\[1\]](#).
2. **Semantic complexity:** Legal abstractions and euphemistic language conceal what data is done (e.g "partners" vs. "third-party advertisers").
3. **Dynamic evasion:** Websites frequently change the method used to grant consent in order to outsmart static auditing tools. (Matte et al., 2020)[\[2\]](#)

This evaluation forms a view towards existing studies on the above challenges through four primary goals:

- Automated extraction of consent dialogues using web scraping techniques
- Detection of dark patterns in interface design through NLP and structural analysis.
- Semantic standardization using frameworks like DPV-2 and ISO 29184 to ensure machine-readable compliance reporting.[\[4\]](#)[\[5\]](#)
- Dynamic automatic real-time compliance monitoring tool development using transformer models.

The purpose of this literature review is to highlight the opportunities to augment systematized GDPR compliance automation, so the system fulfills the identified gaps by analyzing existing work in each area.

2. Review of Relevant Papers

The reviewed relevant papers were selected through a structured search of multiple academic databases which includes IEEE Xplore, Google Scholars, DCU Digital Library, ACM Digital Library, and ArXiv using keywords like 'GDPR compliance,' consent', 'consent dialogues', 'dark patterns', 'semantic standardization', and 'privacy policies'. Out of an initial pool of 20 papers, 6 papers were selected based on their relevance to our research objectives: automated extraction of consent dialogues, detection of dark patterns, semantic standardization using DPV-2 and ISO 29184, and real-time compliance assessment.

2.1 Extraction of Automated Consent Dialogues

This step is essential in understanding GDPR compliance on a large scale: the automated extraction of consent dialogues. In this segment, we critically review existing strategies that have attempted to scrape consent mechanisms and their successes or failures in relation to our aim of building an automated system to scrutinize cookie consent dialogues, including their success and failure.

Dark Patterns after the GDPR: Scraping Consent Pop-ups and Demonstrating their Influence - A Foundation for Web Scraping^[1]

Based on case studies that searched for dark patterns, Nouwens et al. (2020) demonstrate the feasibility of scraping consent management platforms (CMPs), thereby enhancing the foundations for web scraping. Specifically, they successfully scraped 680 dark patterns embedded in the structures of CMPs to measure the level of user consent behaviors. The scraper used structural elements such as the importance of buttons and behavioral elements such as pre-check-user-tick boxes, user control, and other patterns. Surprisingly, only a small percentage of these CMPs succeeded.

In addition, they made technical contributions, such as prioritizing key DOM elements and managing dynamic content with Selenium. For instance, cookie banners were determined to have 3x the scraping priority of other elements. While their technical contributions are valuable, the findings of this research are fairly rigid due to the specificity of the dataset.

The study's focus on UK websites makes the study have a geographic bias and a lack of generalizability to multilingual EU interfaces. As we build your system, it is important to note that this method will not be useful for us to test. Additionally, there is static analysis and a lack of mechanisms to detect A/B tested consent interfaces as well as metadata gaps, which ignores the network-layer consent strings.

Do Cookie Banners Respect my Choice? Measuring Legal Compliance of Banners from IAB Europe's Transparency and Consent Framework - Expanding Methods for Consent String Verification[\[2\]](#)

In contrast, Matte et al. take a different approach by analyzing IAB Europe's Transparency and Consent Framework (TCF). The authors used a Selenium and Chromium based tool called Cookinspect to crawl 1,426 websites and look for consent violations. Additionally, the study is supplemented with a legal examination of GDPR to bridge legal implementations with code.

Their study found that there were serious issues on how consent is collected with issues ranging from consent before any choice and attempts to influence users decision with options. Their effort of analyzing IAB is useful for your research and for how to measure compliance on real websites, the study did not provide a way to conduct a full analysis.

Like Nouwens et al. (2020), the method was limited because it used a static crawler. Further, the study was limited because it only focused on TCF implementations. Our system will address these issues by creating systems to analyze any website with multiple jurisdictions. The semantic drift also only accounts for tracking changes in the scraping cycle, which will be addressed in our project.

2.2 Dark Patterns and the Legal Requirements of Consent Banners: An Interaction Criticism Perspective[\[3\]](#)

Santos et al. (2021) bridge legal, ethical, and human-computer interaction (HCI) perspectives to evaluate how manipulative design practices ("dark patterns") in GDPR-mandated consent banners undermine informed user choice. Their study employs interaction criticism—a methodology combining design analysis, legal interpretation, and user experience evaluation—to dissect consent mechanisms across four dimensions: designer intent, interface properties, user interactions, and societal impact.

The primary purpose of this paper is to identify an intricate legal taxonomy of dark patterns and vividly explain how interface design decisions are made in violation of GDPR regulations. For instance, the authors identify:

- **Tracking walls** (blocking the website access unless users accepts the cookies)
- **Interface interference** (visually prioritizing the "Accept" over the "Reject" buttons)
- **Reduced service models** (degrading functionality for users who reject tracking)

Following patterns directly violate GDPR Article 7's requirement for "freely given" consent. Santos et al. shows us that 45% of analyzed banners used obstruction tactics (e.g., hiding reject options behind multiple clicks), while 43% employed interface interference.

This work is particularly useful for your system's automated dark pattern detection objective. By operationalizing Santos et al.'s taxonomy—for example, flagging skewed button prominence ratios (e.g., "Accept" buttons 2x larger than "Reject")—your NLP models can classify structural violations of GDPR Article 4(11).

2.3 Semantic Standardization

Semantic standardization is a very important component for automating GDPR compliance assessment, as it ensures that consent dialogues and privacy notices can be transformed into machine-readable formats while preserving their legal meaning. Two key contributions in this area are the Data Privacy Vocabulary (DPV) Version 2 by Pandit et al. (2024) and the ISO/IEC 29184:2020 standard for the online privacy notices and consent. These works provides the complementing frameworks for structuring, analyzing, and implementing privacy-related data.

Data Privacy Vocabulary (DPV) Version 2[\[4\]](#)

The DPV, which was created by the W3C Data Privacy Vocabularies and Controls Community Group, provides a machine-readable ontology for various privacy concepts including purposes, legal bases, personal data categories, and technical measures. DPV-2 further develops its predecessor by adding more than 2300 concepts such as `dpv:Consent`, `dpv:hasDataController`, and `dpv:hasPurpose dpv:Marketing`. This taxonomy enables the term ‘third-party partners’, which would be undefined in the GDPR, to be mapped to other defined GDPR-aligned categories.

With DPV-2, as with previous versions, the modular framework allows for jurisdictional extensions (e.g. GDPR, CCPA) and risk management systems adoptions, thus making it friendly to many regulatory environments. For instance, its retention period property, like the rest of it attributes, can also be traced to GDPR Article 5(1)(e) which stipulates data minimization principles. Also, the combination of RDFS+SKOS with OWL2 serializations in DPV-2 enables cost-effective approaches to complex reasoning tasks while retaining advanced models to meet the lightweight deployment requirements.

DPV-2 can be directly applied to our system’s objective concerning semantic uniformity. By incorporating DPV-2 to our scraping pipeline, we can convert disparate consent dialogues into structured formats that can be used on different websites and across various jurisdictions. For example, vague textual descriptions of data recipients can be mapped to `dpv:ThirdParty`, thus enabling automated scrutiny of compliance with GDPR Article 44 pertaining to data transfers within and external to the EU.

Although DPV-2 is particularly useful for unaided domain vocabulary construction, it does not come with web scraping and NLP pipelines integration tools. We bridge this gap by using DPV-2 in combination with BERT-like transformer models that examine both structural and semantic features of consent dialogues.

ISO/IEC 29184:2020[\[5\]](#)

Privacy policies around the world are crafted such that they abide by the regulations and guidelines set forth by ISO/IEC 29184. This document details the control and structure of GDPR and offers protocols for Principles of Transparency (5.2), Principle of Granularity (5.4), and Principle of Accessibility (5.3). These requirements include multi-layer notice design (Section 5.2.7), retention periods declarations (Sections 5.3.11), and disclosures regarding the transfer to third parties (Section 5.3.10).

ISO 29184 is more focused on `dpv:ConsentRecord` as a machine readable record of consent (Annex B). For instance, it becomes necessary to mandate the geo-location of the stored data which is a notice to be generated (Section 5.3.9). This can be done using DPV-2's `epithet` location which improves the interoperability of compliance systems. This assists `dpos` and supports the limitations of `dpv:hasLocation`.

Compliance with the DPV-2 technical features outline stands in regulation in ISO 29184. The reports generated from the system maintain the legal compliance set forth by EU and the industry standard. For example, our browser extension could check compliance regarding whether the website's cookie banner adheres to ISO as layered notice unlike using DPV-2 Windows properties where its content is mapped.

ISO 29184 considers which elements to include in notices and consent mechanisms. However, it does not set any requirements regarding their technical implementation. Bridging this gap in our system involves capturing data through scraping and performing analysis using natural language processing.

2.4 Realtime Compliance Tools[\[6\]](#)

Real-time compliance tools are important to implement because they help automate the assessment of GDPR in contexts where consent mechanisms and privacy policies change frequently. This area was significantly contributed to by Zhang et al. (2024) when they demonstrated how BERT transformer-based models can be used to determine compliance with GDPR Article 13 regulations. Their research focuses on the privacy policies of 5G Mobile Network Operators (MNOs) and assesses compliance via automated classification and readability analytics.

Zhang et al. implemented a BERT-based classification model to determine the adequacy of privacy policies in Article 13 GDPR compliant clauses concerning transparency of data collection, data retention periods, and user rights. The study collected privacy policies from 70 MNOs and annotated them with ten GDPR specific labels such as "Data Retention Period" and "Right to Object to Processing." The model improved performance on most categories, receiving precision and recall scores of over 80%, indicating that credibility content classification was successfully performed.

Along with classification of compliance, the study identifies particular issues in the comprehension of privacy policies. With the Flesch Reading Ease (FRE) and SMOG metrics, the authors pinpoint many policies to be unreasonably sophisticated, which bone fide require a postgraduate understanding in order to digest in full. This is one of the many obstacles that users face in regard to the understanding of their prerogatives under GDPR.

The work of Zhang et al. is useful for our purpose of developing real time compliance tools because of the approach taken. Their application of BERT makes it possible to use transform-based models, meaning GDPR assessments could be automated on a large scale. However, their emphasis on privacy policies as opposed to perpetual consent dialogues poses limitations to the extent their approach can be applied to our system.

Our framework builds on Zhang et al.'s work by changing the focus from dry policy documents to interactive consent dialogues that are performed in real-time, which are more challenging because of how they differ from one site to another. Instead of using predefined classes like Zhang et al. did, our system will first do the scraping, and then, using DPV-2, will standardize meaning on the text in order to classify it. This will enable the system to automatically monitor compliance breaches in consent interfaces.

To illustrate, whereas Zhang et al.'s model would determine whether a particular privacy policy has a section on "Data Retention Period" my system will check whether a cookie banner has some retention period or states "as long as necessary" which is also known as ambiguous retention. The need for this kind of monitoring is critical as it ensures that websites do not adopt and use dynamic avasion tactics.

3. Conclusion

Our analysis of prior works points to an absence of research in the automated evaluation of consent dialogues for GDPR compliance. The web scraping methodologies that were previously employed to capture consent mechanisms have been rather elementary, focusing on static approaches that overlook the dynamic architecture of dark patterns and consent interfaces. By combining monitoring in real-time with the ability to manage multilingual consent dialogues using a language-neutral methodology, our system will solve these challenges.

In addition, our research specifically draws from and expands on previous work by employing DPV-2 and ISO/IEC 29184:2020 as a basis for semantic standardization. With regard to DPV-2, it is beneficial because it serves as an ontology that clusters obscure consent phrases into precise legal terms that are easier to comply with. On the other hand, ISO 29184 offers a permissive context for consent notices as well as the terms of their use, which will be taken into account when creating automated checks for compliance. Therefore, our approach to combining these frameworks makes it possible to close the scene gap between the boundaries set by these policies and efficient technical implementation, making obtaining full surveillance of GDPR compliance easier.

Moreover, translation of consent dialogues will be feasible through the use of transformer-based NLP models drawn from Zhang et al. classifying privacy policies. This will enable us to differentiate between genuine efforts to obtain consent and the use of deceitful dark patterns.

To put it simply, the system will utilize the scraping of websites, semantic normalization, and NLP through transformers in order to develop a sophisticated and customizable compliance checking tool. The findings of this review explain the reasoning behind our selection of technologies, especially DPV-2 and ISO 29184, and aid in formulating strategies for the enhancement of automating GDPR compliance evaluation.

Below is the showing different approaches taken and with the improvements we are trying to make in our system.

Research Objective	Prior Work	Key Contributions	Limitations	Our System's Improvements
1. Consent Dialogue Extraction	Nouwens et al. (2020) Matte et al. (2020)	(i) Web scraping for CMPs (ii) TCF consent string analysis	(i) Static, UK-centric datasets (ii) No dynamic evasion detection	(i) Adaptive scraping (ii) Multilingual support: 24 EU languages
2. Dark Pattern Detection	Santos et al. (2021)	(i) Taxonomy linking dark patterns to GDPR Art. 7 (ii) Interaction criticism framework	(i) Manual classification (ii) No real-time tools	(i) NLP-powered detection: Skewed button ratios (iii) Predictive analytics for emerging patterns
3. Semantic Standardization	DPV-2 (Pandit et al., 2024) ISO 29184	(i) GDPR-aligned ontology (2,300+ concepts) (ii) Compliance metrics for layered notices	(i) No scraping/NLP integration (ii) Static metrics	(i) Hybrid framework: DPV-2 + ISO 29184 mapping (ii) Resolves ambiguous terms like "partners" using machine-readable formats
4. Real-Time Compliance Tools	Zhang et al. (2024)	(i) BERT-based policy classification (80%+ accuracy) (ii) Readability analysis	(i) Focused on static policies (ii) Ignored consent dialogues	(i) Dynamic dialogue analysis: Monitors consent dialogues in real-time (ii) Tracks ambiguous retention terms like "as long as necessary"

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